THE ANH NGUYEN

Graduate Machine Learning Engineer | Computer Vision and Edge AI

Sydney, NSW | mr.bushpro2002@gmail.com | linkedin.com/in/bush2002 | github.com/BushMar12 | bushmar12.github.io

SUMMARY

Graduate machine learning engineer who takes computer vision from dataset to on-device inference in production. In my Master's research, PPO recovery policies I trained raised a PX4 quadrotor's safe-mission rate under rotor failure by 42 points in a pre-registered test. I also built VoiceEye, a live phone app that runs YOLO detection in the browser at about 10 FPS. Master of AI, UTS (GPA 6.67/7, Dean's List 2026). Available now.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

ML and computer vision: PyTorch, scikit-learn, YOLO, RF-DETR, SAM 2, OpenCV, object detection and tracking, Transformers, vision-language models (Qwen3-VL)

RL and robotics simulation: PPO, DQN, Stable-Baselines3, Isaac Sim and Isaac Lab, PX4 software-in-the-loop, MuJoCo, sim-to-real transfer

Deployment and model serving: ONNX Runtime (web and edge), INT8 quantisation, Jetson Nano, Raspberry Pi 5, FastAPI, Docker, PostgreSQL, React

MLOps and cloud: ClearML, Optuna, GitHub Actions CI/CD, AWS (SageMaker Studio), Cloudflare (Pages, Workers AI), Git, Linux, pytest

RESEARCH

What-If Flight Replay: Failure Analysis and Recovery RL for Aerial VLA Policies *2026*

Master's research project, UTS, team of four · co-author on the paper draft · private repository, walkthrough on request

- Team study: each crash of a frozen aerial vision-language-action policy is replayed in MuJoCo with one change against a paired control, and the outcomes become verified labels for a failure monitor and a PPO recovery policy.
- Owned the **Isaac Sim track**: ported the study's procedural city to Isaac Sim and trained PPO policies that recover a quadrotor from rotor failure and wind.
- Built a **PX4 software-in-the-loop study** with fault injection and a pre-registered held-out run of 2,772 missions with zero infrastructure errors.
- Registered result: PPO recovery on top of PX4 raised the safe-mission rate by **42 points over PX4 alone** (95% CI 33 to 52), with the best seed at 88% against 36%.

PROJECTS

VoiceEye: Real-Time AI Vision Assistant *Apr – May 2026*

Computer vision · Edge AI · MLOps github.com/BushMar12/VoiceEye | Live: voiceeye.pages.dev

- Shipped a live phone web app that runs **YOLO26n detection fully on-device at about 10 FPS** with ONNX Runtime Web, built in React and TypeScript on Cloudflare Pages.
- Curated a 48-class dataset from COCO, Open Images and Mapillary Vistas and tuned the detector with Optuna to **0.50 mAP@50** (0.35 mAP@50–95).
- Built a **ClearML pipeline with quality-gated model promotion**, wired into GitHub Actions CI/CD for training and deployment.
- Cut the runtime bundle from 24 MB to about 12 MB, fixed browser memory crashes, and covered the TypeScript tracking pipeline with **144 unit tests**.

JetRacer RL: Sim-to-Real Autonomous Driving (Team Project, UTS) *Sep – Oct 2026*

Reinforcement learning · Robotics · Edge AI Private repository, code walkthrough on request

- Main code contributor. Owned the **obstacle-stop policy, simulator training pipeline and on-car runtime** for a camera-only Jetson Nano car.
- Trained a DQN STOP/GO policy beside the team's PPO steering policy, matched the simulator to the real fisheye camera, and removed a simulator-only shortcut, raising **AUROC on real frames from 0.52 to 0.89**.
- On the real car: **stopped for 14 of 14 objects** with a 20 Hz control loop and 6.9 ms median inference. Set up AWS SageMaker Studio for teammates without a GPU.

Intelligent Vehicle Damage Assessment System (Team Project, UTS) *Apr 2026*

Object detection · Model serving · MLOps github.com/BushMar12/Intelligent-Vehicle-Damage-Assessment

- Team system that detects six damage types, rates severity, estimates repair costs in AUD and writes reports. I maintain the public repository.
- Augmented CarDD from 1,971 to 5,913 images and benchmarked six detectors (RF-DETR-m **76.87% mAP@0.5**). The team deployed YOLO26m: 72.53% mAP@0.5, 42 MB, about 23 ms on GPU.
- Delivered **model serving through a Dockerised FastAPI and PostgreSQL service**, plus ClearML dataset versioning, model promotion and inference monitoring.

Multimodal Deep Learning for Stock Return Prediction (In Progress) *2026*

Deep learning · Time series github.com/BushMar12/Tesla-stock-price-prediction-using-Multimodal

- Two-stream model (Transformer price encoder, 1D CNN news-sentiment encoder, cross-modal attention) for next-day TSLA returns, benchmarked against LSTM, GRU and XGBoost with leakage-safe splits. Results pending.

EXPERIENCE

AI / Computer Vision Intern | FPT Software, Hanoi, Vietnam *Dec 2025 – Feb 2026*

- Reimplemented key **SAM 2** components (prompt encoder, mask decoder, memory encoder, memory attention) in Python to trace how the modules work together in image and video segmentation.
- Prepared input images with LandingAI segmentation and tested how preprocessing and region extraction changed SAM 2 results.
- Ran proof-of-concept segmentation experiments and analysed model behaviour across prompt types and visual inputs.

EDUCATION

Master of Artificial Intelligence | University of Technology Sydney *Feb 2025 – Nov 2026*

Computer Vision sub-major · GPA 6.67/7 · Dean's List 2026, Faculty of Engineering and IT

Bachelor of Engineering in Control and Automation (Advanced Program) | HUST, Hanoi *Sep 2020 – Sep 2024*

AI sub-major · GPA 3.65/4 · **Thesis:** real-time smart parking system with YOLOv8n INT8 on a Raspberry Pi 5, about 30 FPS and 90%+ detection accuracy.

ADDITIONAL INFORMATION

Languages: Vietnamese (native), English (professional working proficiency)